

PYTHON PROGRAMMING LANGUAGE

1. INTRODUCTION TO PYTHON

What is Python?

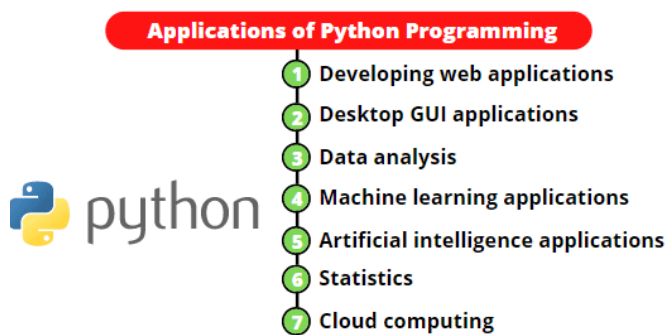
Python is a high-level, interpreted programming language known for its simplicity and readability. It is widely used in various fields, including web development, data analysis, artificial intelligence, scientific computing, automation, and more.

Key Features of Python

1. **Easy to Read and Write:** Python uses a clean and straightforward syntax, making it accessible for beginners.
2. **Interpreted Language:** Python code is executed line by line, which makes debugging easier and allows for interactive programming.
3. **Dynamically Typed:** You don't need to declare variable types explicitly. Python determines the type at runtime.
4. **Extensive Libraries and Frameworks:** Python has a vast collection of libraries and frameworks, like NumPy, pandas, Flask, Django, TensorFlow, and more, which help speed up development.
5. **Cross-Platform:** Python runs on various operating systems, including Windows, macOS, and Linux, making it versatile.

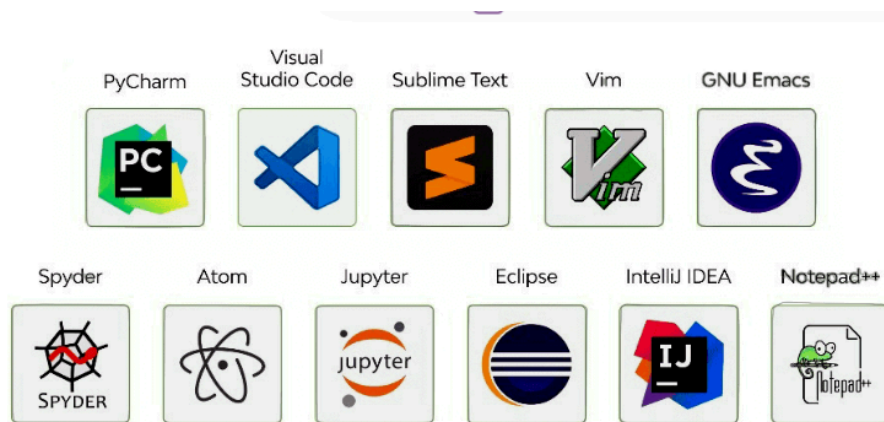
What are the advantages of Python?

Over 125,000 third-party Python libraries exist that enable you to use Python for machine learning, web processing, and even biology. Also, its data-focused libraries like **pandas**, **NumPy**, and **matplotlib** make it very capable of **processing, manipulating, and visualizing data** — which is why it's favored in data analysis.



2. IDLE for python

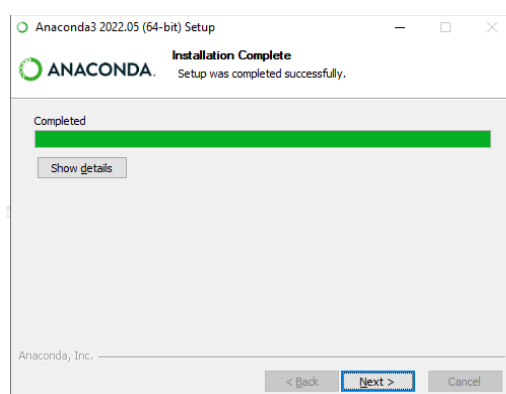
IDLE stands for **I**ntegrated **D**evelopment and **L**earning **E**nvironment and is a free, open-source text editor specifically designed for writing and running Python code



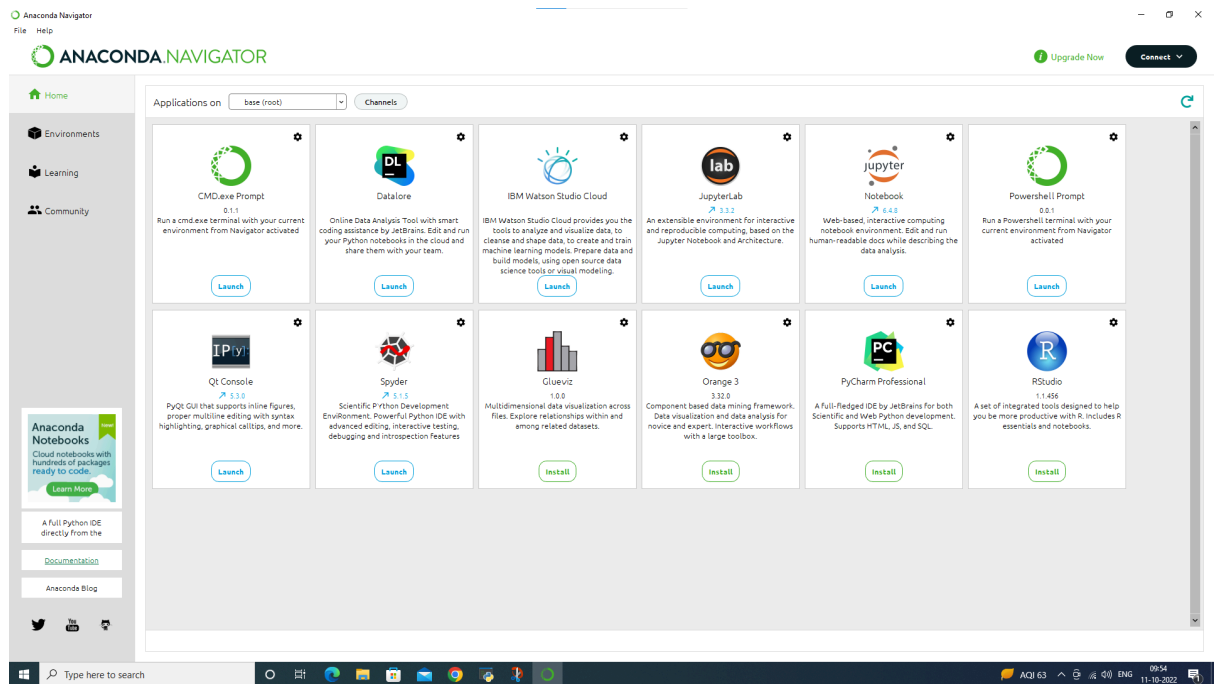
3. DOWNLOAD AND INSTALL PYTHON-ANACONDA

STEPS

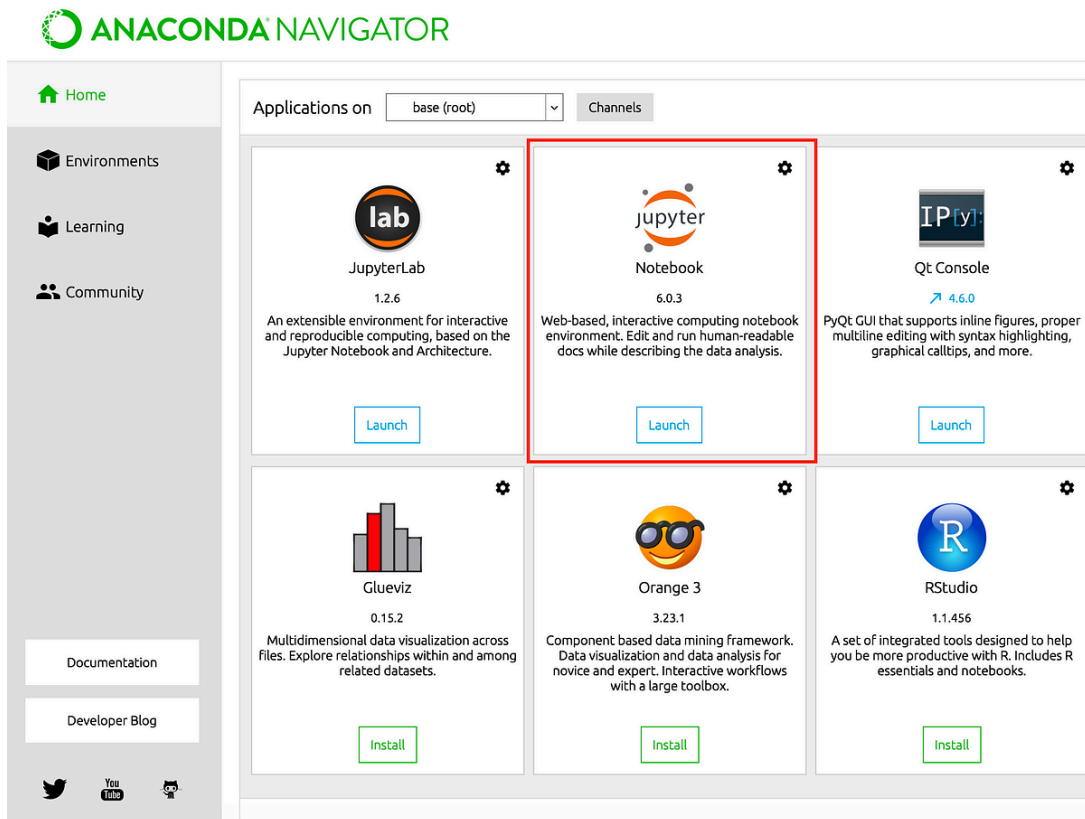
1. Search **Anaconda python download** and download **Anaconda3-2024**
2. From downloads Install Anaconda



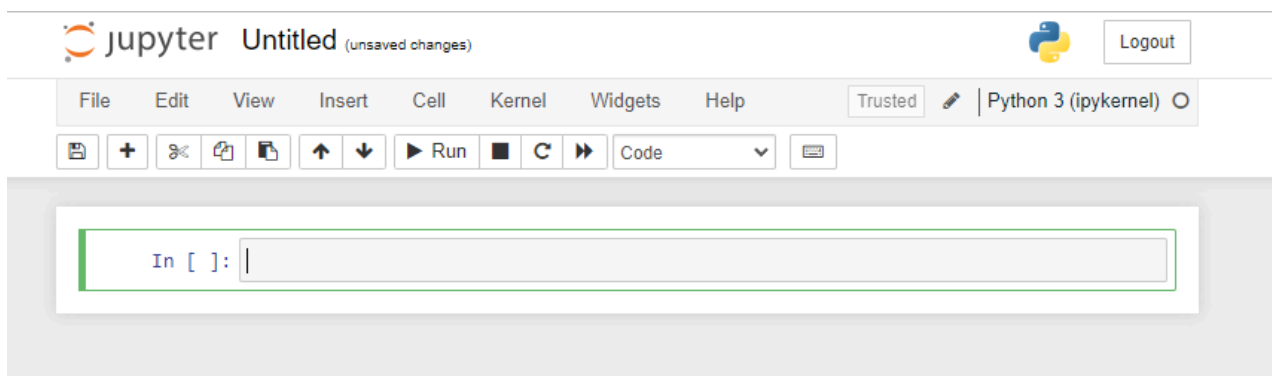
3. Open Anaconda



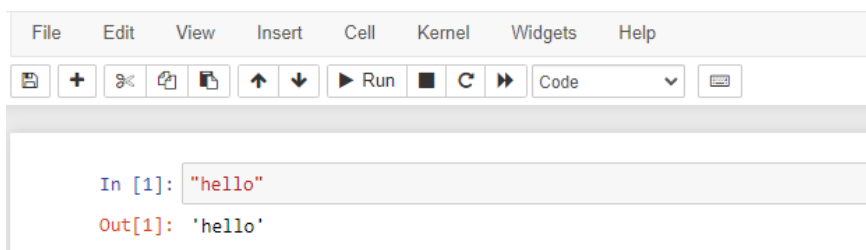
4. Launch Python Jupyter



5. Select New – python 3



4. How to run a program in python

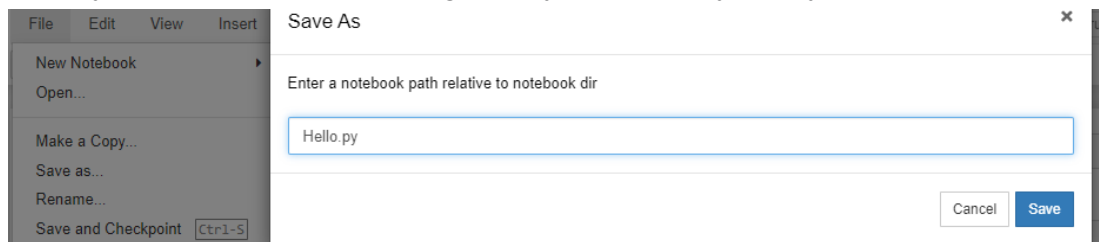


After typing the command in the python cell, we can click on “Run”.

Else **Shift + Enter** run the current cell, select below. Ctrl + Enter run selected cells. Alt + Enter run the current cell, insert below. Ctrl + S save and checkpoint.

5. How to save a python file

Right-click the Python window and select Save As to save your code either as a Python file (.py) or Text file (.txt). If saving to a Python file, only the Python code will be saved.



We can see that the file is saved with extension “.py”

→ **To save in a folder**

.ipynb stands for “Interactive Python Notebook”, the file format of Jupyter Notebook.